Salud360

SALUD 360 · TECHNICAL CHALLENGE · SOFTWARE ARCHITECTURE

COPILOTO SALUD360

Leadership has handed us a mandate. We want to see what you do with it — including the parts you think are wrong.

AT A GLANCE

Format	Take-home, then a 60-minute live session
Your time	Target 4 hours. Please don't spend more.
Length limit	8 pages. We stop reading at 8.
Code required	None
AI tools	Required, and documented

00 READ THIS PART FIRST

This challenge is not a test of how much you can produce. It is a test of judgement, and judgement is easier to see in eight pages than in thirty.

- **Four hours is the target.** If something needs more time than you have, write down what you would do with it and move on — that note is worth more to us than a finished section. Overrunning is not penalised, and it is not credited either. We score what fits in eight pages.
- **Eight pages is a hard cap.** Diagrams count. There is exactly one appendix — the AI-usage appendix in **D8** — and it does not count. Anything else you attach does. If it's page nine, we won't see it.
- **No code.** No repo, no prototype, no working demo. We are hiring an architect, not a weekend.
- **Use AI, and tell us how.** Details in §07. Not using it would be strange; hiding it would be worse.
- **Prose polish is not scored.** Bullets are fine. Spanish is fine if you'd rather — half the panel is Costa Rican, and a Spanish submission scores identically. Write it the way you'd write it for a real design review.

A note on numbering: sections of *this brief* are **§00–§10**. The things we're asking you to produce are **D1–D8**, listed in §06. We'll use those labels in the live session too.

THE ONE THING TO UNDERSTAND

Everything in §04 is the mandate *as it was actually handed to us*. It has not been cleaned up for you. Some of it is right, some of it is expensive, and at least one part of it we should probably refuse to build. Treating the mandate as a specification to implement is the most common way to fail this exercise.

01 WHAT SALUD 360 IS

We are a membership platform for private healthcare in Costa Rica. Members pay a monthly fee — roughly $11 for an individual, $27 for a family, $10 per employee for corporate accounts — and in exchange they get negotiated fixed prices with a network of independent providers: general practitioners, specialists, labs, imaging, dentistry, nutrition, psychology.

We are not an insurer. We do not adjudicate claims. We are closer to a buyers' club with a directory attached. The products a member touches are:

- **The public directory** — search providers by specialty, location, price. Open to anyone, members and prospects alike.
- **The member app** — plan status, dependents, booking, a personal record of consultations.
- **Telemedicine** — an on-demand orientation call with a real clinician, included in the plan.
- **Support** — a human team, by phone and WhatsApp, doing a lot of work that the software above was supposed to be doing.

Providers are independent businesses. They are not our employees and they do not run our software. They agreed to a price list; that is the extent of our leverage.

02 THE PROBLEM, AS THE BUSINESS DESCRIBES IT

From the head of member experience:

> **VERBATIM**
>
> “A member has a sore shoulder for three weeks. They open the directory. They see a list of names, some with prices, some without, sorted by nothing useful. Half of them don't answer the phone. They give up and call us, and one of our agents spends seven minutes doing what the app should have done. Then they tell their HR manager the benefit doesn't work, and we lose the account at renewal.”

From the head of corporate sales: small and mid-size corporate accounts renew at a rate we are not happy with, and the reason employees give in exit interviews is almost always some version of *“I didn't know how to use it.”*

03 THE NUMBERS WE HAVE

Transcribed from three different places: a product dashboard, a spreadsheet our data analyst maintains, and the marketing site. The source of each is noted.

MEASURE	VALUE
Active members billing system	38,000
Providers in the network marketing site	500
Providers in the network provider-database export	748
Provider branches (locations) on record provider-database export	1,140
Branches sharing a phone number with another branch analyst spreadsheet	18%
Providers with any structured availability data analyst spreadsheet	41%
Providers with a complete price list per service analyst spreadsheet	63%
Distinct spellings of specialty names in the database analyst spreadsheet	214
Directory searches per month product dashboard	96,000
Searches that end in a booked appointment product dashboard	4.8%
Support contacts per month, all reasons support dashboard	9,000
Support contacts per month, “where do I go for X” analyst spreadsheet	11,300
Median support handle time support dashboard	6 m 40 s
Fully loaded cost per support contact finance	~$1.85
Gross margin per individual member per month finance	~$3.10

MEASURE	VALUE
Telemedicine consults per month product dashboard	2,400
Telemedicine clinician capacity, consults per month operations, estimate	~3,200
Fully loaded cost per telemedicine consult finance	~$9.40
Corporate renewal rate, accounts under 50 employees sales	71%
Engineers available for this —	4

FAIR WARNING

Some of these figures contradict each other. Some are measured, some are estimates, and we have not labelled which is which. We are not going to tell you which ones to trust. Noticing is part of the exercise.

04 WHAT LEADERSHIP HAS ASKED FOR

These are the five directives as they were given, in the words they were given in.

M1

“A member should be able to describe their symptoms in the app and be told which specialist to see, and book that appointment in the same flow.”

CEO, PRODUCT REVIEW

M2

“Use the best model available and give it our whole provider database in the prompt, so its answers about prices and availability are always current.”

CTO, SAME MEETING

M3

“It has to be live eight weeks from today. I'm presenting to our three largest corporate accounts that same week and I want to show them this.”

CEO

M4

“It cannot add meaningful infrastructure cost. Look at our margin per member.”

CFO

M5

“And it has to work over WhatsApp. That's where our members already are — nobody wants to open another app.”

HEAD OF MEMBER EXPERIENCE

Take this mandate seriously. Then take it apart wherever it deserves to be taken apart, and say so in language you would actually use with the person who said it.

05 WHAT IS GENUINELY FIXED

- **Health data is sensitive personal data under Costa Rican law** (Ley 8968, regulator PRODHAB). Processing it requires express informed consent, and we remain accountable for what any subprocessor does with it. Reason about the shape of this constraint — our counsel owns the letter of it, and we are not asking you to practise law.
- **Providers cannot be forced to adopt anything.** They are independent. We can incentivise, we can make things easy, we cannot mandate.
- **Four engineers and a part-time data analyst.** No ML engineers, no platform team, no SRE on call. The same four people also keep the existing app running.
- **No clinician on staff.** There is a medical advisory board of practising physicians. It meets once a month. The telemedicine clinicians are contracted per consult.
- **The margin is the margin.** $3.10 per individual member per month, and the CFO reads the cloud bill line by line.

06 WHAT TO SEND US

One document, PDF or Markdown, eight pages maximum. D1–D7 count toward the limit; the appendix in D8 does not. Suggested proportions in the right column — they are guidance, not rules, and reallocating them is itself a decision we'll read.

D1	**The answer, up front** What you would build, what you would refuse to build, and the single sentence of reasoning behind each. Written for the CEO, not for us.	~1 pp
D2	**Architecture** One diagram plus the decisions behind it. Be explicit about where a model sits, what it is allowed to be authoritative about, and what it is never allowed to answer on its own.	~2 pp
D3	**The eight weeks** What ships, what gets cut, what exactly the CEO demos on the day. Cutting is the interesting half.	~1 pp
D4	**Unit economics** Cost per member interaction, with the arithmetic shown, including any channel costs. Compare it against a baseline you construct from §03 and state your assumptions — including how much of that baseline you expect to actually displace.	~1 pp
D5	**How you'd know it works** Before launch and after. Include what happens the morning a model version changes underneath you and you didn't ask for it.	~1½ pp
D6	**The version with no AI in it** The cheapest thing that captures most of the value with no model in the loop at all. Then argue honestly about whether your design in D2 actually beats it.	~½ pp
D7	**What this brief doesn't tell you** The questions you'd need answered before committing, ranked by what would change your design, and what you assumed instead in order to keep going. Include the one risk you think could kill this.	~1 pp
D8	**Appendix: how you used AI** See §07. Plus one line: roughly how long this took you. Doesn't count toward the page limit.	any

07 ON USING AI FOR THIS

Use it. We do, all day. Pretending otherwise would tell us something unhelpful about both of us.

In the appendix, tell us:

- Which tools and models, and for which parts.
- One place it was **confidently wrong or usefully shallow**, and how you caught it. This is the part we read most closely.
- One thing you overrode, and why your judgement beat the output.
- One part of this exercise you would never delegate to a model, and what makes it different.
- Roughly how long the whole thing took you. Honestly. It is not scored.

THE RULE THAT MAKES THIS FAIR

In the live session we will ask you to defend specific numbers, claims and trade-offs from your document. If you cannot defend one, it does not count in your favour — regardless of who or what wrote it. A short document you can defend completely beats a thorough one you cannot.

08 THE 60 MINUTES

Three people on our side. Bring your diagram and your own submission — on paper or on screen, whatever you work best from. No new material, no slide deck.

0–10	**You present** Ten minutes for the whole thing. Compression is part of the job.
10–30	**We push** On your assumptions, your numbers, and the parts of the mandate you left alone.
30–50	**The requirements change** We will hand you a new constraint that breaks part of your design, and ask what you do about it. It is not a trick — it is a Tuesday.
50–60	**Your questions** Which we also pay attention to. What you ask about a company tells us what you think matters.

Changing your mind during the session is a positive signal, not a negative one, provided you can say what changed it.

09 WHAT WE'RE ACTUALLY LOOKING AT

Three things, and we score them separately.

One **Critical thinking**

Do you interrogate the problem you were handed, or execute it? Do you notice what is missing, what contradicts itself, and what the stated problem might be a symptom of rather than the cause?

Two **Creativity**

When the obvious answer is blocked by a real constraint, do you find a second and third path? Can you produce something cheaper and stranger than the brief implies — and beat your own proposal with it?

Three **AI knowledge**

Do you know where a language model genuinely earns its place in a system, where it is the wrong instrument, and what it takes to operate one responsibly when the subject matter is somebody's health?

NOT SCORED

Writing polish. English fluency. How attractive the diagram is. Whether you have worked in healthcare before. How many technologies you can name. Whether your conclusion matches ours — there is no single correct architecture here, and the strongest submissions we've seen disagree with each other.

10 SENDING IT IN

- Reply to this thread with the document attached. PDF or Markdown.
- One week from receipt. If you need longer, ask — the answer is almost always yes and it costs you nothing.
- Questions about the brief are welcome and will not count against you. Any question that reveals a real gap in the brief, we'll answer for every candidate.
- We'll schedule the live session within three days of your submission, and we'll tell you who is in the room beforehand.

If any part of this exercise is a barrier for reasons of time, access or accessibility, say so and we will find another way to run it. We would rather adapt the format than lose the candidate.